From Paid APIs to Local GPUs: Zero-Shot Consumer Financial Complaint Classification with Open-Source LLMs

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***Abstract*—Automated classification of consumer financial complaints has significant implications for regulatory oversight and customer service prioritisation. While recent studies have demonstrated strong zero-shot performance using large proprietary language model (LLM) APIs, those approaches incur substantial and recurring monetary costs that render them impractical for resource-constrained researchers and institutions. In this paper we investigate whether open-source, locally deployed LLMs can match or approximate the classification quality of their paid-API counterparts, entirely at zero API cost. We benchmark eight open-weight models (four standard and four reasoning) against the Consumer Financial Protection Bureau (CFPB) dataset, replicating an established five-category zero-shot evaluation protocol on the same dataset. Models are deployed via Ollama on an NVIDIA Tesla V100 GPU within the TRUBA ARF high-performance computing (HPC) cluster, using SLURM for job orchestration. We conduct two experiments: a balanced experiment (200 samples per class, N=1,000) designed for protocol-level comparison with the reference paper, and an imbalanced experiment (N=1,000, preprocessed natural class distribution) to assess real-world robustness. Our best model, `gpt-oss:20b`, achieves a weighted F1 of 0.7451 on the balanced set, exceeding the reported paid GPT-4o score (F1 0.7358) under the matched protocol with zero API spend. Reasoning models form the strongest group on average, although `phi4:14b` is a competitive low-latency standard-model exception. We further document a complete, reproducible HPC inference pipeline and analyse off-label prediction behaviour. Our results demonstrate that competitive zero-shot complaint classification is achievable with open-source infrastructure alone.**

***Index Terms*—consumer complaint classification, large language models, zero-shot learning, open-source LLMs, high-performance computing, CFPB, reasoning models, Ollama, SLURM, natural language processing**

I. Introduction

The Consumer Financial Protection Bureau (CFPB) publishes a public database of consumer complaints submitted against financial institutions in the United States. The dataset spans dozens of product categories and hundreds of thousands of narratives, making it one of the most accessible and practically relevant corpora for natural language processing (NLP) research in the financial domain. Accurately routing and categorising these complaints has direct implications for regulatory efficiency, bank compliance teams, and consumer advocacy organisations: a miscategorised complaint may be forwarded to the wrong department, delayed, or deprioritised [1]–[3].

Unlike many benchmark text-classification datasets, consumer complaint narratives are not merely short topic labels embedded in clean prose. They often contain informal descriptions of stressful financial events, institution names, dates, partial account details, and overlapping product issues. A single complaint may mention credit reporting, debt collection, and a credit card dispute in the same narrative, while the supervised label assigns only one product category. This ambiguity makes the task more realistic than many standard NLP benchmarks and also more valuable as a test bed for instruction-following models: a useful system must map noisy, human-written narratives to a constrained regulatory taxonomy without inventing new labels.

The practical motivation is equally strong. Complaint triage is a high-volume activity in which automation can reduce manual screening time, support earlier detection of recurring institutional problems, and provide consistent first-pass routing. At the same time, financial complaint text is sensitive. Even when datasets are public, a deployable classification workflow for banks, regulators, or research laboratories may need to operate without sending narratives to third-party services. This makes local inference, auditability, and reproducible infrastructure more than implementation details; they are part of the methodological problem.

Traditional approaches to complaint classification relied on bag-of-words representations, support vector machines, and logistic regression. The emergence of BERT [4] and its domain-adapted successors marked a step change, enabling much richer semantic representations and pushing classification accuracy above 90 % in fine-tuned settings [5]. More recently, the advent of instruction-following LLMs such as GPT-3 [6] and its successors opened the possibility of *zero-shot* classification: a model is simply given the candidate categories in a prompt and is asked to assign the most appropriate one, requiring no task-specific training data at all [7], [8].

For consumer complaint analysis, this zero-shot setting is attractive because financial taxonomies evolve and labelled data may be costly to curate for every new jurisdiction, product definition, or institutional workflow. However, zero-shot classification also shifts the main engineering burden from model training to prompt adherence, output parsing, and deployment cost. A model that returns a plausible but unsupported category such as `credit_collection` may look linguistically competent while still failing the regulatory

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routing task. Therefore, evaluating off-label behaviour is essential alongside accuracy and F1.

Roumeliotis et al. [9] conducted a comprehensive zero-shot benchmark of 14 state-of-the-art proprietary and commercial models — including GPT-4o, GPT-4.5, Claude 3.5/3.7, Gemini 2.0, o1, o3-mini, and DeepSeek (API) — on the same CFPB dataset and five-category task. They reported that the best model, OpenAI's o1, achieved an accuracy of 0.777 and an F1 of 0.7764, at a cost of $24.24 per thousand predictions. Their study provided strong evidence that reasoning-enabled LLMs can outperform standard chat models on this structured classification task. However, a critical limitation of their work is that *every* model was accessed through a paid commercial API, creating a barrier to reproduction and deployment in low-budget or data-sensitive environments.

This leaves an important open question: can open-weight models, served locally on shared research infrastructure, deliver competitive zero-shot complaint classification without API access? The answer is not obvious. Smaller local models are cheaper to run but may be weaker at following strict JSON-style instructions. Reasoning models may improve disambiguation and output discipline, but they also generate longer traces and increase latency. In addition, prior balanced benchmarks do not show whether the same ranking survives under the natural class imbalance of the CFPB corpus, where a majority `credit_reporting` class can inflate aggregate metrics.

The present paper addresses this gap directly. Our contributions are:

1) We benchmark eight open-source, open-weight LLMs — `gemma2:9b`, `mistral:7b`, `llama3.1:8b`, `phi4:14b`, `deepseek-r1:8b`, `deepseek-r1:14b`, `qwen3:8b`, and `gpt-oss:20b` — on the same five-category CFPB zero-shot task, at **zero API cost**.
2) We document a complete, reproducible inference pipeline on the TRUBA ARF HPC cluster using Ollama and SLURM, including all deployment parameters needed to replicate our results.
3) We conduct a second experiment using the natural (imbalanced) class distribution, evaluating whether performance patterns observed under artificial balance hold in a more realistic scenario.
4) We provide a direct quantitative comparison with the paid-API results of Roumeliotis et al. [9], showing that our best open-source model exceeds the reported GPT-4o score under the same evaluation protocol and remains competitive with more expensive API-based alternatives.
5) We analyse off-label prediction behaviour — instances where models invent labels outside the five prescribed categories — as a proxy for instruction-following reliability.

The remainder of this paper is structured as follows. Section II surveys the related literature. Section III describes the dataset and preprocessing pipeline. Section IV details the experimental setup. Section V presents results for both experiments. Section VI discusses the findings. Section VII concludes with directions for future work.

II. Related Work

A. From Rule-Based Systems to Transformer Models

The challenge of automatically classifying consumer complaints has a rich history that mirrors the broader trajectory of NLP research. Early systems relied on keyword matching and hand-crafted feature engineering paired with classical classifiers such as support vector machines and naïve Bayes. While effective for narrow, well-defined complaint taxonomies, these approaches proved brittle in the face of the linguistic diversity found in real-world financial complaint corpora [3], [10].

The introduction of BERT [4] triggered a significant paradigm shift. Contextual word embeddings, obtained by pre-training on large general-domain corpora and then fine-tuning on downstream tasks, dramatically reduced the gap between human and machine performance on text classification. Raju et al. [10] applied BERTopic [11], FinBERT, DistilBERT, and RoBERTa to a corpus of 684,653 CFPB complaints, demonstrating that transformer-based topic models could discover latent complaint themes without manual annotation. Vairetti et al. [5] used the Spanish BERT variant BETO for complaint prioritisation, achieving 92.1 % accuracy, illustrating that the BERT paradigm transfers well across languages and regulatory contexts. Das et al. [1] pushed further by coupling transformer-based classifiers with explainability techniques, enabling practitioners to understand *why* a complaint was assigned a given category — a requirement for regulatory compliance. Liu et al. [2] proposed a structured LLM-based categorisation framework (CCF) with real-time monitoring, demonstrating that modern LLMs can serve as end-to-end complaint classification pipelines with minimal domain customisation.

B. The LLM Era and the Zero-Shot Paradigm

The release of GPT-3 [6] introduced the few-shot and zero-shot paradigms, in which a large pre-trained language model can perform novel tasks from a natural language description alone. This capability fundamentally changed the cost calculus of NLP system development: instead of collecting thousands of labelled examples and retraining a model, practitioners could now describe a task in a prompt and obtain reasonable predictions immediately. Zhao et al. [12] provide a comprehensive survey of the capabilities and limitations of modern LLMs, noting that instruction-following performance scales strongly with model size and training quality.

In the financial complaint domain, Correa et al. [13] demonstrated that generative AI can simultaneously classify, summarise, and draft responses to CFPB complaints, achieving 88 % classification accuracy in a zero-shot setting. This result was notable because it required no fine-tuning whatsoever, confirming that the rich language representations encoded in large-scale pre-training are sufficient for this task. Extending the zero-shot paradigm to service industries, Roumeliotis et al. [14] compared GPT-4 Omni and BERT for customer review classification in tourism, finding that GPT-4o achieved 67 % accuracy in a zero-shot setting versus 60.6 % for fine-tuned BERT — an early indication that zero-shot LLMs can challenge fine-tuned specialised models. Li et al. [15] further

showed, through a comprehensive customer service benchmark (TeleEval-CS), that fine-tuned open-source LLMs can match proprietary counterparts when evaluated on domain-specific dialogue tasks, motivating our investigation of open-weight models.

A practically important contextual finding is that the CFPB complaint corpus itself is changing. Shin et al. [16] analysed over 1.1 million CFPB complaints and found that approximately 9.8 % of complaints submitted after the public release of ChatGPT exhibit strong evidence of LLM assistance. Liang et al. [17] independently estimated that roughly 18 % of recent complaints are LLM-assisted. These findings underscore the increasing entanglement of generative AI and financial consumer protection, making robust automated classification even more important.

C. Reasoning Models and the State of the Art

The most recent development in this space has been the introduction of chain-of-thought reasoning models [18], which generate intermediate reasoning steps before arriving at a final answer. DeepSeek-AI [19] demonstrated that open-source models trained with careful scaling laws can match GPT-3.5 tier performance, establishing the viability of open-weight alternatives before the reasoning model era. Building on this foundation, Guo et al. [20] introduced DeepSeek-R1, an open-weight model trained via reinforcement learning to produce structured reasoning traces, achieving performance competitive with proprietary frontier models on a range of benchmarks. Similarly, the Qwen3 series [21] introduced an explicit thinking mode that can be toggled per query.

Roumeliotis et al. [9] brought this reasoning paradigm directly to the complaint classification setting. Testing 14 models on a balanced sample of 1,000 CFPB complaints across five categories, they found that o1 (0.777 accuracy, F1 0.7764) and GPT-4.5 (0.774, F1 0.7763) outperformed their non-reasoning counterparts, and that reasoning models as a class exhibited fewer off-label predictions. Their study is the closest antecedent to the present work and provides the benchmark against which our results are measured.

Concurrently, Bulgakov and Turchin [22] explored DeepSeek-R1-Distill-LLaMA-8B for binary prediction of monetary relief outcomes from CFPB complaints in a zero-shot setting, demonstrating that distilled reasoning models retain competitive performance while fitting on a single consumer-grade GPU. Güllü et al. [23] fine-tuned LLaMA-3.1-8B on the same five-category CFPB task used in our study, reaching 89.99 % accuracy and establishing an approximate fine-tuning performance ceiling for that model family.

D. The Open-Source Gap

Despite these advances, a critical gap remains in the literature. Every paper benchmarking LLMs on CFPB complaint classification has relied either on paid commercial APIs or on fine-tuned models that require substantial labelled training data. Neither approach is accessible to practitioners operating under strict data-governance constraints, limited budgets, or air-gapped computing environments. The present paper fills this gap by restricting itself entirely to open-weight models deployed locally on shared HPC infrastructure, thereby demonstrating that competitive zero-shot complaint classification is achievable without any API cost or proprietary model access.

III. Dataset and Preprocessing

A. Data Source

We use the CFPB Consumer Complaints dataset originally published by Tiwari [24] on Kaggle, which is a processed subset of the public CFPB Consumer Complaint Database. The dataset covers complaints submitted between March 2020 and March 2021 and contains three columns: an unnamed index, `product` (the category label), and `narrative` (the free-text complaint). The raw file comprises 162,421 records consolidated to five target product categories: `retail_banking`, `credit_reporting`, `credit_card`, `mortgages_and_loans`, and `debt_collection`. This is the same dataset and label set used by Roumeliotis et al. [9], ensuring direct comparability.

B. Class Distribution

The original data are heavily imbalanced. After removing records with empty `product` or `narrative` fields (10 rows removed, 162,411 remaining) and discarding narratives exceeding 500 characters (resulting in 103,915 usable records), the class distribution shifts to: `credit_reporting` 63.6 %, `debt_collection` 14.1 %, `mortgages_and_loans` 7.7 %, `credit_card` 7.5 %, and `retail_banking` 7.1 %. Before the length filter, the same five-category corpus contains 162,411 records with `credit_reporting` accounting for 56.1 % of the data.

C. Preprocessing Pipeline

The preprocessing steps, implemented in Python using pandas, are applied in the following order to every record:

1) Drop rows with empty `product` or `narrative`.
2) Retain only records belonging to the five target categories.
3) Remove narratives longer than 500 characters, reducing processing time and GPU memory pressure while avoiding truncation artifacts during inference.
4) Standardise narrative text: convert to lowercase, remove special characters, and normalise whitespace.

D. Experimental Subsets

Two experimental subsets are drawn from the preprocessed corpus.

Experiment A (Balanced): 200 records are sampled uniformly at random from each of the five categories using stratified undersampling with a fixed random seed of 42, yielding $N = 1{,}000$ records. This follows the evaluation protocol of Roumeliotis et al. [9]; because the original sample identifiers are not publicly released, the comparison should be interpreted as protocol-matched rather than instance-identical.

Experiment B (Imbalanced): 1,000 records are sampled using stratified sampling that preserves the post-filter

natural class distribution: `credit_reporting` 636 (63.6 %), `debt_collection` 141 (14.1 %), `mortgages_and_loans` 77 (7.7 %), `credit_card` 75 (7.5 %), and `retail_banking` 71 (7.1 %). This second experiment, absent from the reference paper, assesses how class imbalance affects both accuracy estimates and per-class performance patterns.

IV. Experimental Setup

A. Infrastructure

All inference experiments are conducted on the TRUBA ARF cluster, the Turkish national high-performance computing facility. Jobs are submitted via the SLURM scheduler to the `akya-cuda` partition. Each job is allocated a single NVIDIA Tesla V100-SXM2-16 GB GPU (16 GB VRAM), 10 CPU cores, and a 2-hour wall-clock time limit. Compute nodes have no outbound internet access; all models are pulled in advance on the login node (`arf-ui1`) and stored in the shared `/arf/home` filesystem, which is mounted read-only on compute nodes. This offline-inference configuration mirrors real-world deployments in air-gapped or regulated environments.

Model inference is handled by Ollama v0.30.6, a self-contained serving binary that manages model loading, KV-cache management, and the REST API. The Ollama server is started within the SLURM job, and the Python inference script communicates with it via the local REST endpoint.

B. Models

Table I lists all eight models. Four are standard (non-reasoning) instruction-tuned models; four are reasoning models that generate chain-of-thought traces before producing a final answer. The evaluated Ollama builds use 4-bit quantisation (Q4_K_M GGUF format, Ollama default), which reduces VRAM requirements significantly while maintaining competitive quality. The largest model, `gpt-oss:20b`, requires 13.0 GB VRAM and fits within a single V100. Model names in Table I are the exact Ollama tags used in the experiments. The family column records the upstream lineage or registry metadata associated with each tag and should not be interpreted as equivalence to a proprietary API model with a similar provider name. All artifacts were obtained through `ollama pull <tag>` with no task-specific fine-tuning or local weight modification. `gpt-oss:20b` is an open-weight reasoning model released by OpenAI and distributed through the Ollama registry; it requires no API key and its weights are publicly accessible via `ollama pull gpt-oss:20b`. The "OpenAI" entry in the Family column reflects the model's originating developer, not access through any proprietary API or commercial service.

TABLE I
Open-Source Models Evaluated (Q4_K_M Quantisation Used in This Study)

Model	Family	Type	Params	VRAM (q4)
`gemma2:9b`	Google	Standard	9B	5.4 GB
`mistral:7b`	Mistral	Standard	7B	4.4 GB
`llama3.1:8b`	Meta	Standard	8B	4.9 GB
`phi4:14b`	Microsoft	Standard	14B	9.1 GB
`deepseek-r1:8b`	DeepSeek	Reasoning	8B	5.2 GB
`deepseek-r1:14b`	DeepSeek	Reasoning	14B	9.0 GB
`qwen3:8b`	Alibaba	Reasoning	8B	5.2 GB
`gpt-oss:20b`	OpenAI	Reasoning	20B	13.0 GB

C. Prompt Design

A single, model-agnostic prompt is used for all models and all experiments. The prompt follows the five-label JSON-output template used by Roumeliotis et al. [9] to ensure comparability:

> *You are an AI assistant specializing in consumer complaint classification. Your task is to analyze a consumer complaint and classify it into the most appropriate category from the predefined list: ["retail_banking", "credit_reporting", "credit_card", "mortgages_and_loans", "debt_collection"]. Provide your final classification in the following JSON format without explanations:* `{"product": "chosen_category_name"}`. *Complaint: [NARRATIVE]*

The temperature parameter is set to 0 for all models to reduce sampling variability and improve reproducibility. No explicit `top_p`, `max_tokens`, stop-sequence, or model-specific thinking-mode flags are set; Ollama defaults are used for all other decoding parameters. Although the prompt asks for JSON without explanations, some reasoning-oriented builds may still emit intermediate text before the final answer. The inference script therefore extracts the final JSON object from the response by searching for the last valid `{"product": "..."}` occurrence using a regular expression, accounting for models that prepend reasoning traces before the answer.

The inference script was implemented using only Python standard-library modules (`csv`, `json`, `urllib`, `time`, `argparse`), eliminating the need for any `pip`-installed dependencies on the compute node. Each request is retried once on HTTP failure before being recorded as an error; errors are included in the final count and do not silently inflate accuracy. The raw model response is retained alongside the parsed label in the output CSV to allow post-hoc auditing of all parsing decisions. The `gpt-oss:20b` model required a 4-hour SLURM wall-clock limit due to its larger size; all other models completed inference within the standard 2-hour limit.

D. Evaluation Metrics

Following the reference paper, we compute four primary metrics over all N=1,000 predictions: *exact-match accuracy*, *weighted precision*, *weighted recall*, and *weighted F1-score*. We additionally report *macro-F1*, which weights all classes equally and is especially informative for the imbalanced experiment. Weighted variants weight each class by its sample size, making them appropriate for comparison with the reference paper but potentially optimistic under class imbalance. We additionally record the *mean latency* (seconds per prediction, wall-clock time) and the number of *off-label* predictions — responses where the model returned a category name not in

TABLE II
EXPERIMENT A RESULTS — BALANCED ($N = 1{,}000$)

Model	Type	Acc.	Prec.	Rec.	F1	Macro-F1	Lat. (s)	OL
gpt-oss:20b	R	0.745	0.764	0.745	0.745	0.745	2.59	0
deepseek-r1:14b	R	0.716	0.733	0.716	0.716	0.716	5.92	0
phi4:14b	S	0.709	0.736	0.709	0.710	0.710	0.55	1
qwen3:8b	R	0.707	0.739	0.707	0.707	0.707	3.89	0
gemma2:9b	S	0.676	0.717	0.676	0.673	0.673	0.66	1
deepseek-r1:8b	R	0.675	0.709	0.675	0.674	0.674	4.64	1
mistral:7b	S	0.596	0.676	0.596	0.584	0.584	0.24	54
llama3.1:8b	S	0.587	0.697	0.587	0.596	0.596	0.57	57

R=Reasoning, S=Standard, OL=Off-label, Lat.=Mean latency

the prescribed five, or failed to return valid JSON entirely. Off-label predictions are counted as errors. The API cost for all experiments is $0. Compute time is supplied by an institutional HPC allocation rather than a commercial LLM API.

V. RESULTS

A. Experiment A: Balanced Evaluation

Table II presents the full results for Experiment A. The gpt-oss:20b reasoning model achieves the highest accuracy (0.7450) and F1 (0.7451), followed by deepseek-r1:14b (F1 0.7162) and the standard model phi4:14b (F1 0.7095). Three reasoning models occupy the top four positions (gpt-oss:20b rank 1, deepseek-r1:14b rank 2, qwen3:8b rank 4), with the standard phi4:14b ranked third. Notably, deepseek-r1:8b (rank 6) falls below the standard gemma2:9b (rank 5), indicating that model size interacts with the reasoning benefit. The two weakest models are mistral:7b (F1 0.5842) and llama3.1:8b (F1 0.5958), both of which also produced the highest off-label counts (54 and 57, respectively).

The OL column in Table II is particularly informative. The three zero-off-label models (gpt-oss:20b, deepseek-r1:14b, and qwen3:8b) are all reasoning models, whereas the two models with severe format-following failures are standard models: llama3.1:8b produces 57 off-label predictions and mistral:7b produces 54. Aggregated by model type, the reasoning group attains a mean F1 of 0.710 with only one off-label prediction across 4,000 outputs, while the standard group attains a mean F1 of 0.641 with 113 off-label predictions. This makes instruction adherence, not only classification accuracy, a central part of the performance gap. At the same time, phi4:14b is an important exception among standard models: it reaches F1 0.7095 with only one off-label prediction and a mean latency of 0.55 s, making it the strongest low-latency alternative.

Because all models are evaluated on the same 1,000 balanced examples, we also perform post-hoc exact McNemar tests on paired correctness outcomes for the leading model. The advantage of gpt-oss:20b is statistically significant against deepseek-r1:14b ($p = 0.0106$), phi4:14b ($p = 0.0030$), qwen3:8b ($p = 0.0008$), and gemma2:9b ($p < 10^{-7}$). All four comparisons remain below a Bonferroni-adjusted threshold of $\alpha = 0.0125$. These tests support the ranking observed in Table II, while still leaving multi-seed

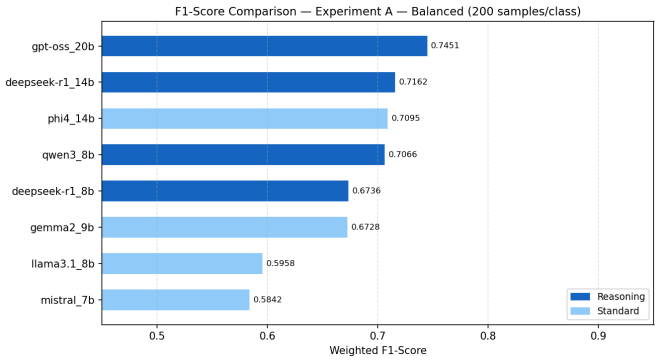


Fig. 1. Weighted F1-scores for all models in Experiment A (Balanced). Reasoning models are shown in dark blue; standard models in light blue.

resampling as an important direction for future work. For example, gpt-oss:20b is correct on 88 instances where phi4:14b is incorrect, while the reverse occurs on 52 instances; the corresponding counts are 75 versus 46 for deepseek-r1:14b, 81 versus 43 for qwen3:8b, and 115 versus 46 for gemma2:9b. By contrast, adjacent high-performing models are not clearly separated: phi4:14b is correct on 71 instances where qwen3:8b is incorrect, while the reverse occurs on 69 instances ($p = 0.9327$). The corresponding comparisons are also non-significant for deepseek-r1:14b versus phi4:14b (76 versus 69, $p = 0.6184$) and deepseek-r1:14b versus qwen3:8b (67 versus 58, $p = 0.4744$), indicating that their rank ordering should be interpreted cautiously.

For Experiment A, macro-F1 is numerically identical to weighted F1 because all five classes have equal support. Per-class inspection nonetheless reveals that even the best model is uneven across labels: gpt-oss:20b reaches recall 0.885 on credit_reporting but only 0.625 on debt_collection. This motivates the macro-F1 reporting in Experiment B, where minority-class behaviour can otherwise be hidden by the majority class. Detailed per-class precision, recall, and F1 for the leading models are reported in Table V.

Fig. 1 visualises the F1-score distribution across models. Reasoning models (dark blue) cluster at the top while standard models show greater spread. The confusion matrix for the best-performing model (gpt-oss:20b) is shown in Fig. 2. The model handles credit_reporting and retail_banking well but shows some confusion between mortgages_and_loans and credit_card. Fig. 3 shows the confusion matrix for phi4:14b, the best standard model: its strong diagonal confirms reliable five-class classification with only one off-label prediction, and minor confusion between mortgages_and_loans and credit_card. The confusion matrix for llama3.1:8b (Fig. 4) illustrates the impact of heavy off-label prediction: a large fraction of samples are assigned invented categories, degrading performance across all five classes.

B. Experiment B: Imbalanced Evaluation

Table III presents results for Experiment B. All eight models show numerically higher accuracy and F1 compared to Experiment A. gpt-oss:20b again leads with accuracy 0.8040 and

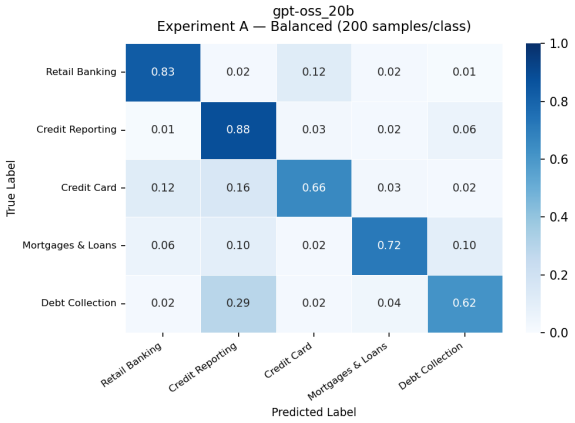


Fig. 2. Confusion matrix for `gpt-oss:20b` (best model, Experiment A). Rows are true labels; columns are predicted labels.

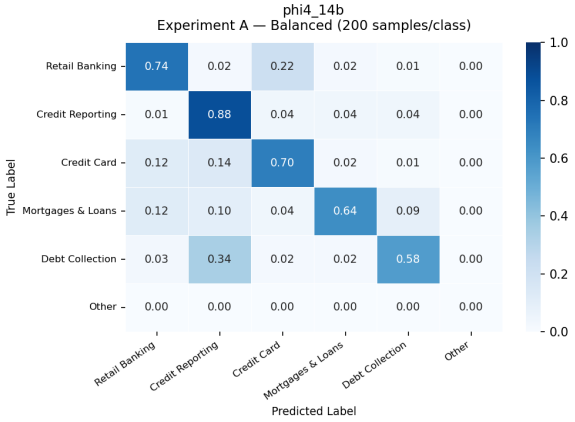


Fig. 3. Confusion matrix for `phi4:14b` (best standard model, Experiment A). Strong diagonal with minor confusion between `mortgages_and_loans` and `credit_card`, and only one off-label prediction.

F1 0.8042. Notably, the standard model `phi4:14b` reaches a near-identical score (accuracy 0.7980, F1 0.7987), while `qwen3:8b` also matches it (accuracy 0.7980, F1 0.7950). The table therefore shows that the imbalanced setting changes the absolute score level more than the identity of the leading models: `gpt-oss:20b`, `phi4:14b`, `qwen3:8b`, and `deepseek-r1:14b` remain the strongest group. However, the large gains for `llama3.1:8b` and `mistral:7b` should not be interpreted as genuine robustness improvements, because both models still produce many off-label outputs and benefit from the dominant `credit_reporting` class.

The relative model ranking is broadly preserved compared to Experiment A. However, as discussed in Section VI, the improvement in accuracy scores must be interpreted carefully in light of the dominant `credit_reporting` class. The added macro-F1 column makes this clearer: `gpt-oss:20b` has weighted F1 0.8042 but macro-F1 0.7237, while `llama3.1:8b` has weighted F1 0.7092 but macro-F1 0.5555. The gap shows that weighted metrics substantially overstate minority-class performance in the imbalanced setting. The F1-score distribution across all Experiment B models mirrors Experiment A in its ranking (Fig. 5 right bars), with `gpt-oss:20b`, `phi4:14b`, and `qwen3:8b` again forming

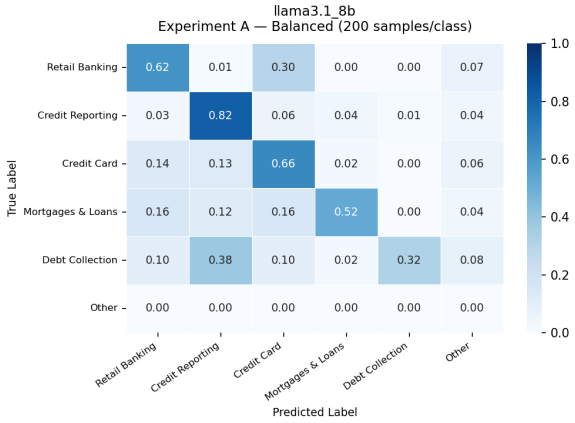


Fig. 4. Confusion matrix for `llama3.1:8b` (worst model, Experiment A). The high off-label rate (57 predictions) manifests as diffuse error across all classes.

TABLE III
EXPERIMENT B RESULTS — IMBALANCED ($N = 1{,}000$)

Model	Type	Acc.	Prec.	Rec.	F1	Macro-F1	Lat. (s)	OL
`gpt-oss:20b`	R	0.804	0.806	0.804	0.804	0.724	2.39	0
`phi4:14b`	S	0.798	0.803	0.798	0.799	0.700	0.54	1
`qwen3:8b`	R	0.798	0.799	0.798	0.795	0.703	3.60	0
`deepseek-r1:14b`	R	0.790	0.802	0.790	0.794	0.702	5.72	0
`deepseek-r1:8b`	R	0.771	0.777	0.771	0.768	0.663	4.41	4
`gemma2:9b`	S	0.762	0.785	0.762	0.762	0.661	0.66	0
`llama3.1:8b`	S	0.687	0.782	0.687	0.709	0.556	0.56	49
`mistral:7b`	S	0.686	0.777	0.686	0.703	0.536	0.23	36

R=Reasoning, S=Standard, OL=Off-label, Lat.=Mean latency

the leading group.

C. Cross-Experiment Comparison

Fig. 5 provides a side-by-side view of accuracy across both experiments. All models improve numerically from Experiment A to Experiment B; the shift is most pronounced for `llama3.1:8b` (Δacc $= +0.100$) and `mistral:7b` (Δacc $= +0.090$), the two models with the highest off-label counts. In terms of weighted F1 the corresponding gains are even larger ($+0.113$ and $+0.119$ respectively). This pattern suggests that under imbalanced conditions, a model defaulting to the majority class can still achieve a deceptively high aggregate score.

D. Comparison with Paid-API Models

Fig. 6 compares all eight of our open-source models with the five best reported results from Roumeliotis et al. [9]. Table IV summarises the key comparison points. Our top model, `gpt-oss:20b`, achieves a higher F1 (0.7451) than the GPT-4o score reported in the reference benchmark (0.7358) while requiring no API expenditure. `phi4:14b` and `qwen3:8b` both approach the GPT-4o performance tier. Because the reference paper does not publish the exact sampled complaint identifiers, this comparison should be read as protocol-matched rather than paired on identical instances. The gap to o1 (F1 0.7764) and GPT-4.5 (F1 0.7763) remains non-trivial, suggesting that the

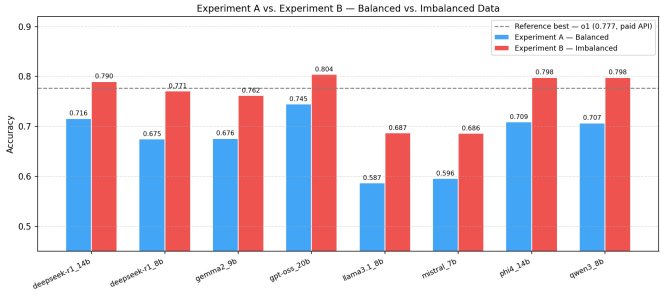


Fig. 5. Side-by-side comparison of accuracy across Experiment A (Balanced, blue) and Experiment B (Imbalanced, red) for all models. The dashed reference line is labelled o1 (0.777), the best paid-API result from the reference paper. The upward shift in Experiment B partially reflects class imbalance rather than genuine model improvement.

TABLE IV
COMPARISON WITH REFERENCE PAPER (BALANCED, $N = 1{,}000$). PAID-API RESULTS SOURCED FROM ROUMELIOTIS ET AL. [9].

Model	Source	Accuracy	F1	Cost/1k
o1	Paid API	0.777	0.7764	$24.24
GPT-4.5	Paid API	0.774	0.7763	$12.32
Claude 3.7 Sonnet	Paid API	0.765	0.7626	$0.64
Gemini 2.0 Flash	Paid API	0.758	0.7586	$0.04
GPT-4o	Paid API	0.736	0.7358	$0.45
`gpt-oss:20b`	Local (ours)	0.745	0.7451	$0.00
`deepseek-r1:14b`	Local (ours)	0.716	0.7162	$0.00
`phi4:14b`	Local (ours)	0.709	0.7095	$0.00
`qwen3:8b`	Local (ours)	0.707	0.7066	$0.00

reasoning depth of frontier proprietary models still provides an edge for this structured task. Nevertheless, the cost differential is stark: the reference paper reported o1 costs of $24.24 per thousand predictions, versus $0.00 for all our models. Thus, Table IV supports a more nuanced claim than simple "open-source wins" or "paid APIs win": the best local model does not reach the strongest paid reasoning models, but it does exceed the reported GPT-4o baseline and removes recurring API cost entirely. This is the central deployment trade-off of the paper. For low-budget, privacy-sensitive, or offline environments, `gpt-oss:20b` offers a competitive zero-shot option; for settings where the last few F1 points matter more than cost or data locality, frontier paid reasoning models still retain an advantage.

E. Latency Analysis

Fig. 7 plots accuracy against mean prediction latency for all models in Experiment A. Standard models are notably faster: `mistral:7b` completes a prediction in 0.24 s on average, while `phi4:14b` requires only 0.55 s despite being twice the parameter count. Reasoning models are slower by one to two orders of magnitude relative to the smallest standard models: `deepseek-r1:14b` takes 5.92 s per prediction due to its chain-of-thought trace. The latency–accuracy trade-off is therefore sharply model-type dependent: `phi4:14b` achieves

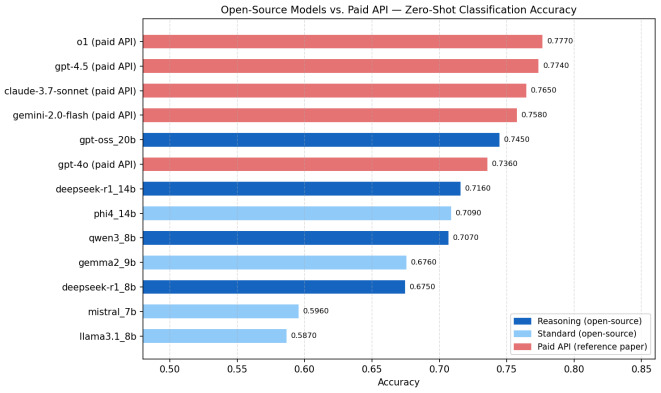


Fig. 6. Comparison of all eight open-source models (Experiment A) against five paid-API baselines from Roumeliotis et al. [9]. Horizontal bars show accuracy.

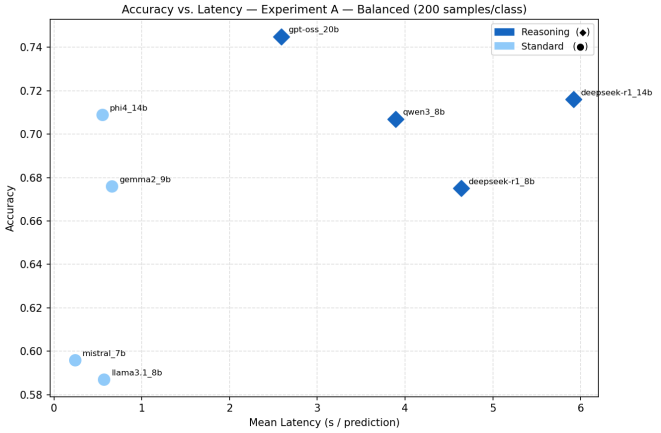


Fig. 7. Accuracy vs. mean prediction latency (seconds) for all models in Experiment A. Reasoning models (diamonds) tend to be slower but more accurate; standard models (circles) vary widely. `phi4:14b` achieves the best accuracy-to-latency ratio.

the best accuracy-to-latency ratio among all models, requiring just 0.55 s per prediction while reaching F1 0.7095.

F. Per-Class Breakdown for Leading Models

Table V reports per-class F1 for the three leading models in each experiment, confirming that aggregate weighted F1 masks important minority-class variation. In Experiment B all three models jump to $\geq$0.87 F1 on `credit_reporting` while minority classes remain in the 0.57–0.76 range, confirming the majority-class bias effect discussed in Section VI.

TABLE V
PER-CLASS F1 FOR LEADING MODELS (EXP. A AND B)

E	Model	Retail	CreditRep.	Card	Mortg.	Debt
A	`gpt-oss:20b`	0.8098	0.7195	0.7154	0.7902	0.6906
A	`phi4:14b`	0.7259	0.7054	0.6998	0.7457	0.6705
A	`qwen3:8b`	0.7700	0.6870	0.6613	0.7740	0.6409
B	`gpt-oss:20b`	0.7771	0.8810	0.6316	0.7179	0.6109
B	`phi4:14b`	0.7006	0.8831	0.6182	0.6624	0.6378
B	`qwen3:8b`	0.7595	0.8801	0.5750	0.7006	0.5975

E = Experiment; Mortg. = Mortgages & Loans.

VI. Discussion

A. Reasoning vs. Standard Models

The most consistent finding across both experiments is that reasoning models dominate the upper performance tier. Three of the four reasoning models appear in the top four positions of Experiment A (`gpt-oss:20b`, `deepseek-r1:14b`, `qwen3:8b`), with the standard `phi4:14b` placed third. This pattern aligns with the findings of Roumeliotis et al. [9], who similarly found that o1 and o3-mini outperformed GPT-4o and other non-reasoning counterparts, and with the broader evidence that chain-of-thought prompting elicits more reliable structured outputs from LLMs [18], [20]. The underlying mechanism is plausibly that reasoning allows the model to decompose ambiguous narratives before committing to a label, reducing noise-driven misclassifications.

However, the advantage is not universal. `deepseek-r1:8b` (Reasoning, 8B parameters, F1 0.6736) is outperformed by `phi4:14b` (Standard, 14B, F1 0.7095) and is comparable to `gemma2:9b` (Standard, 9B, F1 0.6728). This suggests that *model scale* interacts strongly with the reasoning benefit: at 8B parameters, reasoning distillation may not provide enough representational capacity to overcome the advantage of a larger, high-quality standard model.

B. Off-Label Prediction Behaviour

Off-label predictions — responses where the model returned a label not in the prescribed five categories — are a direct measure of instruction-following reliability. The two models with the worst F1 scores (`llama3.1:8b`: 57; `mistral:7b`: 54) also exhibit the highest off-label counts in Experiment A. This is not coincidental: off-label responses are counted as prediction errors, directly reducing accuracy and recall. In Experiment A, all four reasoning models produce at most one off-label prediction each, and `gpt-oss:20b`, `deepseek-r1:14b`, and `qwen3:8b` produce zero. The better instruction-following of reasoning models reflects the structured nature of chain-of-thought: the model explicitly reasons about which label is appropriate before formatting its output, reducing the likelihood of hallucinating a novel category.

The raw status logs show that off-label behaviour has two distinct forms. `llama3.1:8b` primarily fails at the formatting/parsing level, producing 57 parse errors in Experiment A and 49 in Experiment B. In contrast, `mistral:7b` usually returns parseable JSON but invents labels outside the allowed taxonomy, producing 54 valid-but-out-of-vocabulary labels in Experiment A and 36 in Experiment B. Smaller counts for `phi4:14b`, `gemma2:9b`, and `deepseek-r1:8b` are also valid-but-out-of-vocabulary labels such as `unknown` or `health_insurance`. This distinction matters operationally: parse errors indicate output-format failure, whereas out-of-vocabulary labels indicate failure to stay within the regulatory taxonomy.

Notably, the off-label counts for `mistral:7b` and `llama3.1:8b` improve substantially in Experiment B (36 and 49 respectively, compared to 54 and 57 in Experiment A). This may reflect a simpler classification decision under imbalanced conditions, where the dominant `credit_reporting` class provides an easier default that is more likely to be formatted correctly.

C. Latency–Accuracy Trade-off

From a practical deployment perspective, the latency results present an important consideration. The fastest model, `mistral:7b` (0.24 s), achieves only F1 0.5842 — the worst result in Experiment A. `phi4:14b`, at 0.55 s per prediction and F1 0.710, offers the most favourable accuracy-to-latency ratio of any model tested. For applications requiring both real-time response and reasonable accuracy, `phi4:14b` is the recommended choice.

Reasoning models are consistently slower: `deepseek-r1:14b` takes 5.92 s per prediction. At 1,000 predictions, this translates to roughly 1.6 hours of total inference time on a single V100. For large-scale regulatory applications processing millions of complaints, the time budget would need to be considered alongside model quality. All reported latency figures for reasoning models include the full chain-of-thought generation trace; the final JSON label is extracted from the end of this trace via the regex described in Section IV. Latency comparisons between reasoning and standard models should therefore be interpreted as end-to-end wall-clock cost rather than pure decoding speed.

D. Interpreting the Imbalanced Results

The across-the-board increase in accuracy and F1 from Experiment A to Experiment B demands careful interpretation. While a genuine improvement is possible — models may be aided by the greater frequency of `credit_reporting` training signal in their pre-training corpora, which coincides with its frequency in the CFPB data — a more parsimonious explanation exists: *class imbalance inflates aggregate accuracy metrics*.

When 63.6 % of the test set belongs to a single class, a model that classifies most ambiguous narratives as `credit_reporting` will achieve a high weighted accuracy even if its per-class performance is poor. The particularly pronounced gains for `llama3.1:8b` (+11.3 F1 points) and `mistral:7b` (+11.9 F1 points) — the two highest off-label producers in Experiment A — are consistent with this mechanism: under imbalanced conditions, a confused model benefits disproportionately from majority-class guesses. We therefore caution against using Experiment B accuracy as a simple indicator of model quality and recommend that future evaluations report per-class precision and recall alongside aggregate metrics.

E. Comparison with the Reference Paper

Our results confirm that competitive zero-shot classification is achievable with open-source, locally deployed models. The `gpt-oss:20b` model exceeds the reported GPT-4o benchmark on both accuracy and F1 under the matched five-category protocol, while incurring no API fee. This is a meaningful result: GPT-4o costs $0.45 per thousand predictions in the reference benchmark, meaning that at moderate scale

(one million complaints per year), local deployment would avoid roughly $450 in API fees while keeping classification quality in the same performance tier. This should not be read as computation being free: local inference still consumes GPU time, electricity, system administration effort, and institutional HPC capacity. The practical claim is therefore about eliminating recurring third-party API charges and data transfer, not about eliminating all operational cost.

The remaining gap to the frontier paid models (o1, GPT-4.5, Claude 3.7) is approximately 3–4 F1 points. Whether this gap is consequential depends on the application. For initial triage and routing, an F1 around 0.74–0.75 may be fully adequate. For compliance-critical decisions, the gap warrants attention — though it could plausibly be closed by fine-tuning [15], [23] or ensemble methods.

F. Limitations

This study has several limitations. First, the balanced comparison matches the reference protocol but not necessarily the exact sampled instances, since the original sample identifiers are unavailable. Second, the 500-character length filter reduces inference cost but also shifts the class distribution, especially increasing the share of `credit_reporting`; future work should test whether the same patterns hold for longer narratives. Third, lowercasing and special- character removal may discard useful cues such as account-number patterns, capitalization, dates, or punctuation; the effect of less aggressive LLM-oriented preprocessing remains untested. Fourth, all experiments use $N{=}1{,}000$ examples, one random seed, and a five-class taxonomy; larger temporal slices, multiple resampled test sets, and richer taxonomies may reveal additional drift and rarer category confusions. Fifth, all local models are evaluated with a single zero-shot prompt. Prompt wording, output-format instructions, few-shot examples, or self-consistency decoding may change the relative ranking of models, particularly for smaller standard models with weaker instruction following. Sixth, all local models are evaluated in their q4 Ollama quantised form, so the reported scores conflate model family, model size, and quantisation effects. Seventh, we cannot rule out pre-training contamination: open-weight models may have encountered CFPB complaints or derivative datasets during pre-training, although no task-specific fine-tuning is used in our experiments. Eighth, the reported $0 cost refers to API expenditure, not to the full economic cost of GPU hardware, electricity, or HPC operation. Finally, the main tables report weighted aggregate metrics; for operational deployment, macro-F1 and per-class precision/recall should be inspected to avoid masking poor minority-class performance.

G. Infrastructure Contribution

A secondary contribution of this work is the documented HPC deployment pipeline. The combination of SLURM job scheduling, offline Ollama serving, and a standard-library-only Python inference script provides a template that can be applied to other NLP classification tasks on shared research clusters without requiring pip-installed dependencies on the compute node. This reproducibility infrastructure lowers the barrier for other researchers to benchmark local LLMs on domain-specific tasks.

VII. Conclusion

We have presented a comprehensive zero-shot benchmark of eight open-source large language models on the CFPB consumer financial complaint classification task, conducted entirely at zero API cost on the TRUBA ARF HPC cluster. Our results demonstrate that the best open-source model, `gpt-oss:20b`, achieves a weighted F1 of 0.7451 on a balanced evaluation, exceeding the reported paid GPT-4o score (F1 0.7358) from the reference paper by Roumeliotis et al. [9] under a matched protocol. Reasoning models generally occupy the strongest performance tier, although the large standard model `phi4:14b` (F1 0.7095) offers the best latency–accuracy trade-off among all models tested.

Our secondary imbalanced experiment reveals important evaluation dynamics: aggregate accuracy improvements under imbalanced conditions are partially attributable to majority-class bias rather than genuine model improvement, underscoring the need for per-class reporting in real-world settings.

From a practical standpoint, these findings validate the viability of self-hosted LLM deployments for financial NLP tasks: institutions that cannot share sensitive complaint text with third-party API providers, or that operate under budget constraints, can now achieve competitive classification quality using open-weight models on standard GPU hardware.

Future work will explore five directions. First, a *quantisation sensitivity study* will evaluate all eight models at q4, q8, and fp16 precision to determine whether the accuracy gains from higher numerical precision justify the increased VRAM cost — a sub-study not attempted in the reference paper. Second, a *prompt and sampling sensitivity study* will repeat the balanced and imbalanced evaluations across multiple random seeds and prompt variants, reporting confidence intervals and paired significance tests beyond the post-hoc McNemar analysis presented here. Third, and most directly motivated by the present results, we will *fine-tune the same eight open-weight models* on the CFPB training split and compare their supervised performance against their zero-shot counterparts reported here. This paired comparison will quantify the exact cost of removing the annotation requirement and clarify whether the 3–4 point F1 gap to frontier paid APIs can be closed by task-specific supervised training. Fourth, *long-narrative robustness* will be examined by relaxing the 500-character filter and measuring whether additional complaint context improves classification or instead increases prompt length and latency costs. Fifth, the benchmark will be *extended to the full multi-year CFPB corpus* with a richer category taxonomy to assess scalability and temporal robustness of the open-source inference pipeline.

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Conflict of Interest

The authors declare that they have no conflict of interest.

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Code Availability

All source code, preprocessing scripts, and SLURM job files used in this study are publicly available at: https://github.com/muhammettan28/ConsumerComplaint

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